

Heart Disease Classification Using Machine Learning Models

Prepared by:

Masoumeh Ostvar

Department of Computer Engineering

Winter 2025

Contents

Classification Evaluation Metrics	2
1. Logistic Regression Model.....	2
2. Support Vector Machine (SVM) Model	3
3. K-Nearest Neighbors (KNN) Model	4
Performance Comparison of the Models	5

Classification Evaluation Metrics

To analyze and compare the performance of the classification models, the following evaluation metrics are used:

- **Accuracy:** The ratio of correctly predicted instances to the total number of instances.
- **Precision:** The ratio of correctly predicted positive instances to the total number of instances predicted as positive.
- **Recall (Sensitivity):** The ratio of correctly predicted positive instances to the total number of actual positive instances.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced measure of both metrics.
- **Confusion Matrix:** A table that summarizes the number of correct and incorrect predictions made by the model for each class.

1. Logistic Regression Model

Logistic Regression				
Accuracy: 0.5489130434782609				
Classification Report:				
	precision	recall	f1-score	support
0	0.69	0.91	0.79	75
1	0.46	0.50	0.48	54
2	0.18	0.08	0.11	25
3	0.27	0.15	0.20	26
4	0.00	0.00	0.00	4
accuracy			0.55	184
macro avg	0.32	0.33	0.31	184
weighted avg	0.48	0.55	0.50	184

Accuracy: Moderate (approximately 55%).

Precision and Recall:

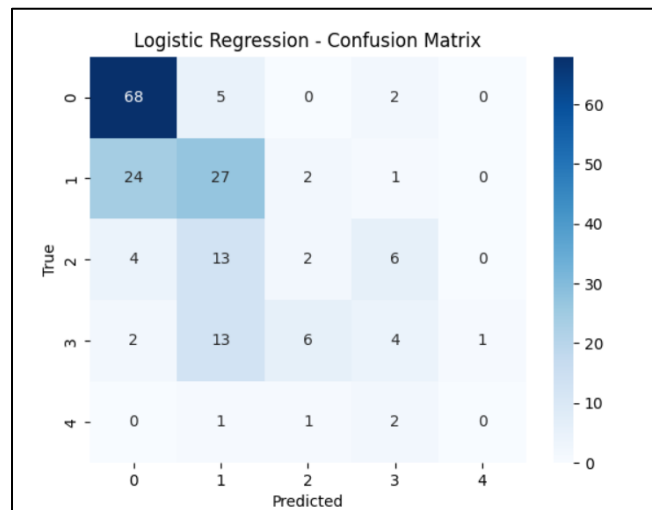
Class 0 (No Heart Disease) has relatively high precision (approximately 69%) and high recall (approximately 91%).

Classes 1 to 4 (different stages of heart disease) show lower precision and recall values, particularly Class 4, which has a precision of zero.

F1-Score:

Class 0 demonstrates good performance, while the other classes achieve relatively low F1-Scores.

Confusion Matrix:



The model correctly predicts most samples belonging to Class 0. However, Classes 2, 3, and 4 are rarely identified correctly.

The Logistic Regression model performs well in identifying healthy individuals (Class 0), but it shows poor performance in distinguishing different stages of heart disease. Most samples belonging to Classes 1, 2, and 3 are incorrectly classified as Class 0. Therefore, this model appears to be more suitable for binary classification problems rather than multi-class disease severity classification.

2. Support Vector Machine (SVM) Model

```
Support Vector Machine
Accuracy: 0.5217391304347826
Classification Report:

```

	precision	recall	f1-score	support
0	0.65	0.91	0.76	75
1	0.42	0.46	0.44	54
2	0.00	0.00	0.00	25
3	0.20	0.12	0.15	26
4	1.00	0.00	0.00	4
accuracy			0.52	184
macro avg	0.45	0.30	0.27	184
weighted avg	0.44	0.52	0.46	184

Accuracy: Moderate (approximately 52%).

Precision and Recall:

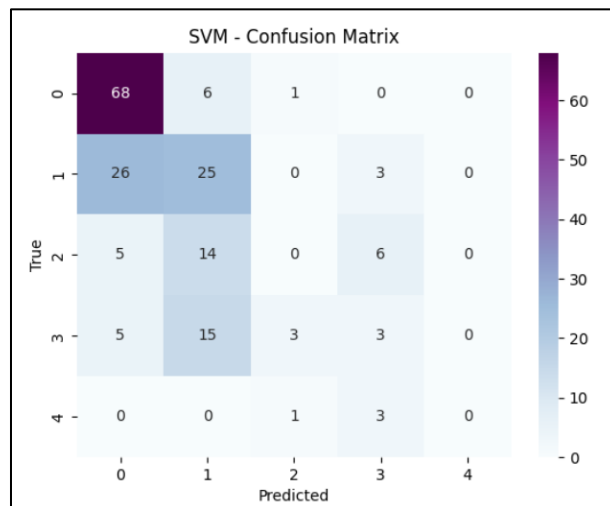
Class 0 (No Heart Disease) has relatively high precision (approximately 65%) and high recall (approximately 91%).

Classes 1 to 4 exhibit lower precision and recall values, particularly Classes 2 and 4, which have a precision of zero.

F1-Score:

Class 0 demonstrates good performance, while the remaining classes achieve relatively low F1-Scores.

Confusion Matrix:



The model correctly predicts most samples belonging to Class 0. However, Classes 2 and 4 are rarely identified correctly.

The SVM model demonstrates performance similar to Logistic Regression, although it performs slightly worse overall. Most samples from Classes 2 and 4 are incorrectly classified as Class 0. Therefore, while SVM is generally well suited for high-dimensional datasets, it does not achieve satisfactory performance on this particular dataset.

3. K-Nearest Neighbors (KNN) Model

```
K-Nearest Neighbors
Accuracy: 0.5271739130434783
Classification Report:
```

	precision	recall	f1-score	support
0	0.64	0.91	0.75	75
1	0.40	0.43	0.41	54
2	0.36	0.16	0.22	25
3	0.25	0.08	0.12	26
4	1.00	0.00	0.00	4
accuracy			0.53	184
macro avg	0.53	0.31	0.30	184
weighted avg	0.48	0.53	0.47	184

Accuracy: Moderate (approximately 52%).

Precision and Recall:

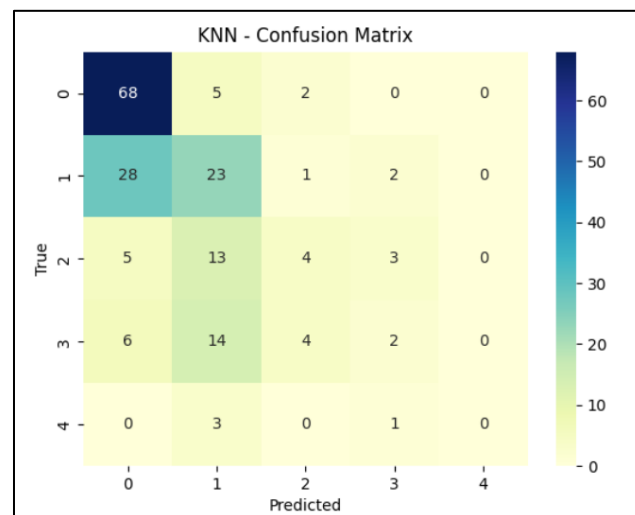
Class 0 (No Heart Disease) has relatively high precision (approximately 63%) and high recall (approximately 85%).

Classes 1 to 4 exhibit lower precision and recall values, particularly Class 4, which has a precision of zero.

F1-Score:

Class 0 demonstrates good performance, while the remaining classes achieve relatively low F1-Scores.

Confusion Matrix:



The model correctly predicted most samples belonging to Class 0. However, Classes 2, 3, and 4 were not identified accurately.

The KNN model showed performance similar to SVM and Logistic Regression, but it performed slightly better on Classes 2 and 3. Nevertheless, most samples from Class 4 were incorrectly classified into other classes. Therefore, it can be concluded that although KNN is generally more suitable for smaller datasets, it did not achieve satisfactory performance on this dataset.

Performance Comparison of the Models

Model	Accuracy	Precision (avg)	Recall (avg)	F1-Score (avg)
Logistic Regression	55%	48%	55%	50%
SVM	52%	42%	52%	46%
KNN	52%	45%	52%	47%

Model	Class 0(No Disease)	Class 1	Class 2	Class 3	Class 4
Logistic Regression	Highest Number of Correct Predictions	Many Incorrect Predictions	Almost Not Recognized	Many Incorrect Predictions	Almost Not Recognized

SVM	Many Correct Predictions	Many Incorrect Predictions	Almost Not Recognized	Many Incorrect Predictions	Almost Not Recognized
KNN	Fewer Correct Predictions than the Other Two Models	Many Incorrect Predictions	Slightly Better than SVM	Slightly Better than SVM	Almost Not Recognized

Among the evaluated models, Logistic Regression achieved the best overall performance; however, it remained limited in accurately identifying the different stages of heart disease. SVM and KNN demonstrated similar performance, with slightly better results for Classes 2 and 3. All models struggled to correctly classify Classes 2, 3, and 4, suggesting that class imbalance in the dataset may have contributed to these difficulties.